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Task Description:

Understanding of qualitative phenomena based on experiments supported by accurate computational simulations (steady and unsteady) for quasi-steady internal condensing flows in micro-gravity and terrestrial environments is essential for good design and performance of condensers in applications such as Rankine-cycle based power (nuclear) plants, Thermal Management systems, Looped Heat Pipes, Capillary Pumped Loops, etc.
Condenser-sections in these devices often achieve complete or incomplete condensation by the exit under unspecified or specified (as being proposed) exit conditions (which are exit vapor quality or liquid fraction or pressure) while the inlet (pressure and flow rate) and wall thermal conditions are either known or specified through a conjugate problem. These condensing flows' attainability/stability of steady flows, intrinsic waviness, noise-sensitivity (primarily due to minuscule transverse vibrations of the condensing surface), and exit-condition sensitivity critically depend on the presence or absence of gravity in driving the condensate.

The new understanding reported here show how shear driven 0g flows are fundamentally different than the ubiquitous gravity driven and gravity dominated flows. The 0g shear driven internal condensing flows can be achieved with repeatability only if they are realized in a carefully designed set-up with adequate flow control strategies and vibration isolation that fix steady inlet conditions (mass flow rate and inlet pressure) and, for any given steady or transient heat load, ensures that the exit-condition (exit pressure or exit quality) is "actively" or "passively" kept within a certain "desired range" described in this report. The simulation results and experimental results for shear driven partial condensation in horizontal channel (which is shown to be equivalent to 0g flows if the vapor flow is only partially condensed, i.e. up to a certain amount – e.g. 60-90% of the inlet flow rate) show that annular/stratified condensing flows in zero-gravity are achievable. However these flows are structurally unstable under ordinary arrangements for the operation of the condenser that do not specify and steady a correct number of boundary conditions. Structural instability of the annular/stratified flow could occur either through unplanned variability in the number of boundary conditions that are specified while one of the boundary conditions is being adjusted ("actively" or "passively") or it could occur through specification of the exit boundary condition outside a "desired" range. These results indicate that, for shear driven zero gravity flows, without the recommended flow control strategies and vibration isolation, the natural tendency of the condensate is to select from any number of complex liquid-vapor morphologies depending upon the nature of time varying inlet, exit, and condensing-surface thermal conditions that inadvertently gets imposed by a conventional system design (which can still work very well for gravity assisted flows). Therefore both the reported results on the inherent difficulties (i.e. "elliptic-sensitivity" of the 0g shear driven flows make them sensitive to exit pressure impositions in the presence of fluctuations, noise-sensitivity, geometry sensitivity, and variability in the operating conditions due to large thermal-transients induced by "elliptic-sensitivity") and the reported remedies (reported flow control strategies as well as the ones under development) to achieve repeatable quasi-steady shear driven condensing flows (fully or partially condensing flows) need to be given due attention in any proper space-based system design.

The ground based experiments (both 0g simulating horizontal channel experiments and its differences from vertical tube experimental results) and supporting simulations (both for terrestrial and micro-gravity environments) show how flow control strategies (that includes control of exit-condition effects) can be used to attain repeatable and desirable quasi-steady flows in space and thus avoid unwanted unsteadiness and oscillatory flows as well as avoid or reverse unwanted thermal-transients. Without the recommended control, the unsteady/oscillatory flows and unwanted thermal-transients may arise from changes in flow regime within the condenser or from system-instabilities (due to interaction of condenser with other upstream or downstream components of the system).

Although the reported research results provide fundamental understanding of flows inside relatively simpler tube and channel geometries, the qualitative understandings/results and tools that have been obtained are sufficiently fundamental and general in nature that they will provide an important framework for facilitating the design, incorporation, and operation of new realistic space-based or low-gravity condensers (that is condensers that employ more complex but realistic geometries) and boilers (by suitable extensions of the reported results) in planned systems. In fact the basic result regarding a need for independent control for removing unwanted fluctuations and instabilities arising from inadvertent non-natural pressure-difference imposition (even when they are minuscule and in the range of 5-50 Pa) is general enough that novel flow control strategies must be developed.

Task Objectives

Experimental

• Experimental investigation of attainability/stability and exit-condition issues for downward and steady internal condensing flows inside a vertical tube. Similar experimental investigations of shear driven internal condensing flows in a horizontal channel (i.e. rectangular cross-section of 2 mm height) - which experience condensation on its bottom wall - has been initiated and partially completed. The key qualitative results from these experiments have been shown to be in agreement with the qualitative results obtained from 0g simulations for a geometry that is close to the experimental geometry.

• Ensure that the ground based experimental facility is well instrumented and progressively capable of providing more and more detailed information that can be synthesized with results obtained from the computational capability being developed for these flows.

• We have fully reported an invention and development of a non-intrusive film thickness sensor (based on principles of fluorescence and fiber-optics) that is capable of measuring time-varying thickness of the dynamic condensate. When additional funding becomes available, we plan to make progress on employing such sensors in the reported ongoing horizontal channel . experiments.

Computational

• Complete new developments for earlier reported simulation tool for laminar vapor/laminar condensate internal condensing flows for channel or vertical tube geometries.

• Ensure that the simulation tool accurately locates the interface (under smooth or wavy conditions) for terrestrial and zero-gravity environments and meets various convergence and grid independence criteria for both specified and unspecified exit condition cases.

• Ensure that the simulation tool is capable of reliably reproducing known results for classical benchmark problems (such as Nusselt problem of film condensation on a vertical plate, the Koh problem for forced vapor flow experiencing film condensation on a horizontal plate, etc.) and, also, capable of providing new non-linear stability results consistent

with known experimental results for these classical problems.

• The verifications of the two dimensional code on classical problems as well as through successful comparisons of the simulation results with the reported ground based experiments have been reported here. Further development and verification of this tool with results obtained from ISS experiments is planned and this work is in progress.

• Eventually the simulation tool should be reliable enough to help reduce the number of design iterations needed to have a condenser (or boiler) operate successfully in a zero gravity environment.

Rationale for HRP Directed Research:

Issue-1: Can one achieve, in zero or micro-gravity, an internal condensing flow in which the condensate adheres to the condensing surface? Can one do this even by the conventional approach, which says that the flow is solely determined by the inlet and the wall heat-transfer conditions? Can an approach be developed to make the condensate adhere to the condensing surface in the desirable regimes (such as annular/stratified regime with or without a preceding mist-flow regime) for shear driven flows in 0g? Flows with predominantly annular/stratified regime (and insignificant plug/slug and bubbly regimes) are desirable for ensuring efficiency, repeatability, and functionality of space-based condensers. Research Result: Yes, such flows can be achieved, but only with the help of the new understanding reported here on how shear driven 0g flows exhibit "elliptic-sensitivity" and are fundamentally different than the ubiquitous gravity driven and gravity dominated flows. The 0g shear driven internal condensing flows can be achieved with repeatability only if they are realized in a carefully designed set-up with adequate flow control strategies and vibration isolation that fix steady inlet conditions (mass flow rate and inlet pressure) and, for any given steady or transient heat load, ensures that the exit-condition (exit pressure or exit quality) is “actively” or “passively” kept within a certain “desired range” described in this report. The simulation results and experimental results for shear driven partial condensation in horizontal channel (which is shown to be equivalent to 0g flows if the vapor flow is only partially condensed, i.e. up to a certain amount – e.g. 60-90% of the inlet flow rate) show that annular/stratified condensing flows in zero-gravity are achievable. However these flows are structurally unstable under ordinary arrangements for the operation of the condenser that do not specify and steady a correct number of boundary conditions. Structural instability of the annular/stratified flow could occur either through unplanned variability in the number of boundary conditions that are specified while one of the boundary conditions that is being adjusted (“actively” or “passively”) is outside a “desired” range for steady realization of the flow. These results indicate that, shear driven zero gravity flows exhibit both "parabolic" and "elliptic" sensitivities, and without the recommended flow control strategies and vibration isolation, the natural tendency of the condensate is to select from any number of complex liquid-vapor morphologies and thermal boundary-conditions depending upon the nature of the cooling methodology and quasi-steady inlet and exit conditions that may inadvertently get imposed by a conventional system design (which can still work very well if one was dealing with gravity assisted flows). Therefore both the reported results on the inherent “elliptic-sensitivity” of the shear driven flows and the reported flow control strategies to achieve repeatable annular/stratified shear driven condensing flows (preferably partially condensing flows), need to be given due attention in any proper space-based system design.

Issue-2: Can one use experiments and simulations to synthesize results for gravity and shear driven flows to better understand the similarity and differences between space-based shear driven flows and ground-based gravity driven flows? Can such an understanding yield better ability to incorporate a condenser in a specified gravity environment (on moon, in 0g, etc.) or in a mode where its performance is independent of the magnitude and orientation of the gravity vector?

Research Impact/Earth Benefits:

Research Result: Yes, the research presents a synthesis of experimental results with the help of a computationally/theoretically developed transition map between shear driven 0g flows and gravity dominated flows. The reported results for shear driven horizontal channel experiments make predictions that are trustworthy aids in the design of space based and 0g experiments. The results also allow one to estimate performance in any non-zero gravitational environment (moon, etc.) because of our use of a certain non-dimensional gravity parameter in the theoretically/computationally obtained transition maps and correlations.

The only gravity-insensitive operation of a condenser that is shown to be possible involves use of a stack of micro-m scale hydraulic diameter ducts and its operation under suitable conditions. These are part of our ongoing non-NASA investigation plans as they are more relevant to modern terrestrial applications (such as electronic cooling).

Issue-3: Can the reported use of ground based horizontal channel (gap height of 2 mm) and reported 0g steady and unsteady simulations replace ISS flight experiments? If not, what information is needed from flight experiments?

Research Result: No, the research presents two results. The first demonstrates a need for flight experiments and the second demonstrates the utility of the ongoing ground-based experiments. There is a reliable steady simulation result presented here which shows that the flow downstream of the location where a horizontal channel annular/stratified condensing flow transitions to plug/slug flow (under an arrangement which allows self-seeking exit-condition) is significantly affected by the presence or absence of transverse gravity. Therefore ground-based horizontal channel experimental results for these downstream locations will not be the same for 0g flows where the thickening film is not exposed to the transverse weight of the condensate. The second result shows that upstream of the smaller of the two lengths (where transition to plug/slug flows would occur in the presence or absence of transverse gravity), the ground based horizontal channel experiments and space based experiments are expected to yield nearly identical results.

This transition location for horizontal channel ground-based flows corresponds to a transition from annular/stratified flows to plug/slug flows. A similar but different location - and perhaps a different kind of transition - is expected for 0g flows at a farther downstream distance where the vapor shear will be low and the liquid flow would again be ready to encapsulate and carry stagnant vapor bubbles or plugs. New enhanced unsteady 0g simulation capabilities for identifying this length of the condenser prior to flight experiments as well as synthesis of such results with flight experiments are needed but are currently unavailable. Continuation of this research is also needed to identify this length – particularly so if one needs to plan condenser operations, which avoid more complex vapor/liquid morphologies and associated problems with vapor flow control.

Task Progress:	The results reported here establish the following (with the help of a synthesis of experiments and simulations): (i) Stable and steady shear driven 0g annular/stratified flows are theoretically and experimentally shown to be possible under steady inlet and condensing-surface thermal conditions if one of the boundary conditions (say the exit pressure) is free to adjust and is not exposed to constraints or strong fluctuations arising from downstream components. This is due to "parabolic" and "elliptic" sensitivities of these flows. (ii) The stability of annular/stratified 0g condensing flows is weak and the flows are noise-sensitive to condensing-surface vibrations as well as to fluctuations in the boundary conditions (this includes exit conditions). However suitable flow-control strategies can make these flows more robust. (iii) We have developed and reported two independent but mutually supportive simulation tools that provide: (a) a qualitative transition map between shear driven and gravity driven flows, and (b) some useful quantitative correlations for 0g flows realized under "natural/parabolic" conditions. (iv) We have experimentally demonstrated repeatable attainability of quasi-steady "parabolic" flows in a horizontal channel. Up to the reported points of transition from annular/stratified flows to plug/slug flows (which can not be avoided for fully condensing flows), these experimental flows and 0g flows are computationally shown to be the same. A flow regime map for flow in horizontal channel and a separate but different flow regime map for channel flows in 0g is possible and can be developed if this research is continued. (v) We have experimentally and theoretically demonstrated the hitherto unidentified phenomenon of "elliptic-sensitivity" of these shear driven 0g flows. This phenomenon comes in to play because the flows, in the presence of fluctuations, respond to imposition of elliptic boundary conditions by changing the mean flow variables - including heat-transfer rates and thermal-transients induced changes in the condensing-surface temperatures. (vi) Most importantly we are able to report and plan flow control strategies that can allow stable operation of space-based condensers despite its weak stability as well as its various fluctuation and boundary-condition sensitivities. However this is possible only if we move away from ordinary conventional system designs to designs that exploit the above reported experimental and computational results - which show a way to do this. Through a proper exploitation of the above research results for the operation of a flow condenser in space and the integration of these results with results obtained from other ongoing research results for flow boilers and liquid/vapor separators, more efficient and controlled thermal systems (which includes space based Rankine cycle, vapor compression refrigeration cycles, air-conditioners, etc.) can be developed and their functionality in space can be ensured.
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